

BANARAS HINDU UNIVERSITY

B.Sc. (Hons) Semester IV Examination 2025-26 (NEP)

CHEMISTRY**Paper Code: CHEMJ44 : Techniques in Chemistry**

Ref. No.: Conf/Even/N/2025-26/09

Time: 2½ Hours

Maximum Marks: 70

Instructions: Answer five questions including Question 1 which is compulsory. The figures in the right-hand margin indicate marks.

Question 1.

- (a) What is a standard solution? Write the difference between primary and secondary standard solutions with suitable examples. [2]
- (b) What is an indicator? Describe its different types with suitable examples. [2]
- (c) Sketch the various types of bending vibration in IR of H₂O. [2]
- (d) Why pH ranges from 0 to 14 only? [2]
- (e) Calculate the pH value of 10⁻⁹ mole HCl. [2]
- (f) What is crystallisation? State its importance in purification. [2]
- (g) What is the difference between qualitative and quantitative analysis? Explain with examples. [2]

Question 2.

- (a) Describe Instrumentation of UV-Visible absorption spectroscopy and its applications. [4]
- (b) Explain the concepts of molarity, normality and molality. Derive their expressions and discuss their interrelationships with suitable numerical examples. [5]
- (c) Explain the principle of complexometric titration. Describe the role of EDTA, metal ion indicators and applications of this method. [5]

Question 3.

- (a) Discuss the process of crystallization and precipitation. Explain the factors affecting crystal formation, purity and yield, along with their applications in chemical purification. [6]
- (b) Discuss the concept of conjugate acid-base pairs. [2]
- (c) Explain the general concepts of acids and bases according to Arrhenius, Brønsted-Lowry and Lewis theories, with suitable examples. [6]

Question 4.

- (a) Define the following terms used in UV-Visible absorption spectroscopy, with suitable examples: [6]
- (i) Bathochromic shift
 - (ii) Hypsochromic shift
 - (iii) Hypochromic effect
 - (iv) Hyperchromic effect
- (b) Discuss the principles and procedures of acid-base titrations. Explain the titration curves for a strong acid vs. a strong base and a weak base vs. a strong acid. [4]
- (c) A 25 μL serum sample was analyzed for glucose content and found to contain 26.7 μg . Calculate the concentration of glucose in $\mu\text{g/ml}$. [4]

Question 5.

- (a) Describe the principle of chromatography. Compare paper chromatography, thin-layer chromatography (TLC) and column chromatography in terms of method, advantages and applications. [8]
- (b) What is the Lambert-Beer law? Explain each term in the expression. [2]
- (c) Write short note on chemical shift in NMR. [4]

Question 6.

- (a) Describe infrared (IR) spectroscopy. Explain different IR sources. How we prepare sample for analysis? [5]
- (b) Describe redox titrations, including their principles, types and applications. [5]
- (c) Write notes on pH and pH calculations. [4]

Question 7.

- (a) Explain the principles of distillation. Describe in detail steam distillation and vacuum distillation, including their working mechanisms, advantages and applications. [8]
- (b) Differentiate between the fingerprint region and the group frequency region in IR spectroscopy. [3]
- (c) Explain precipitation titrations in details. [3]
